V4 power and wiring: battery, regulator and schematic

The Mercury V1 sends the servo position signal; a small 2S battery and a 5 V regulator power the SG90. Part sizes below come from the sellers' listings I could reach; I could **not** read the Mercury manual (the Altimeter Cloud site is not reachable from this session), so its servo-header pin order and signal voltage are the items to confirm.

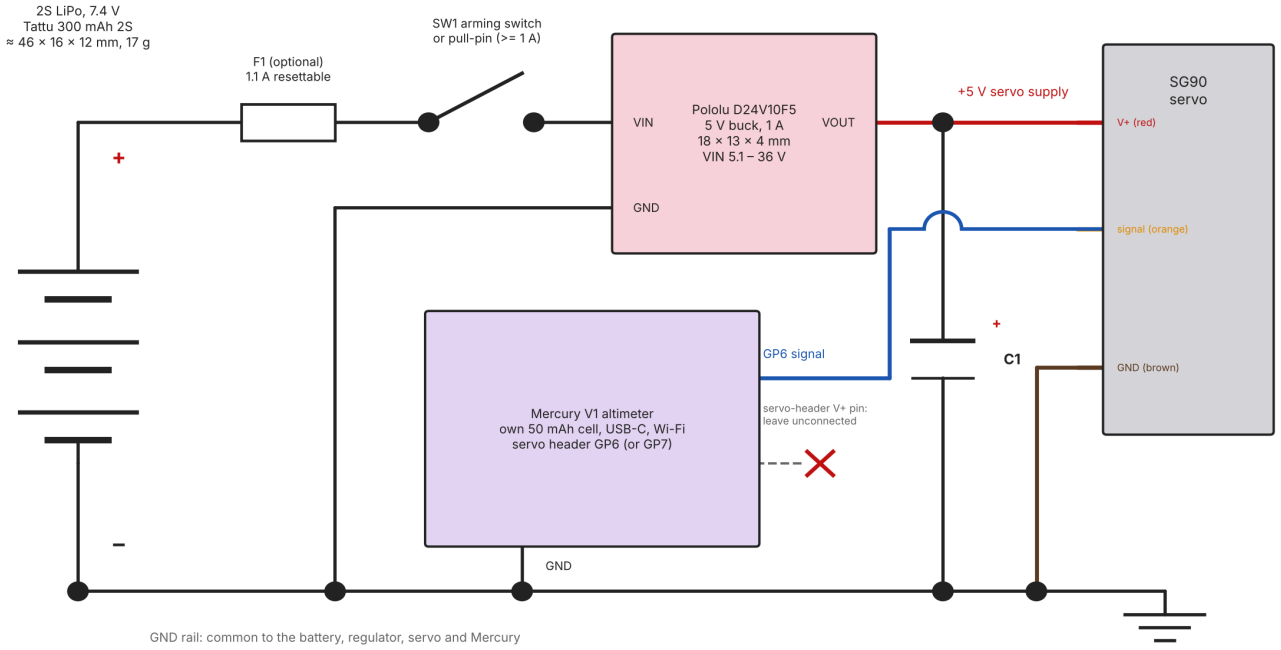
1 Recommended parts

Part	Recommendation	Size / weight (from listings)	Why
Battery	**Tattu 300 mAh 2S, 75C** (HV, 7.6 V nominal, JST-PHR plug)	45–47 × 16–17 × 12–13 mm, 16–17 g (sellers differ by about 2 mm)	Small, light, high-rate 2S pack; 7.4–8.7 V in, which the regulator below accepts; 300 mAh is far more than a flight needs
	Alternative: Turnigy nano-tech 2S 300 mAh 35C	44 × 12 × 17 mm, 17 g	Thinner, but listed as discontinued
Regulator	**Pololu D24V10F5**: 5 V, 1 A step-down	0.5 × 0.7 × 0.14 in (about 13 × 18 × 4 mm), 1 g	Input 5.1–36 V; the SG90 takes 4.8–6 V and draws roughly 0.3–0.55 A at stall (sources vary), so 1 A is about 2× headroom
	Option for more torque: D24V10F6 (6 V, 1 A)	same footprint	SG90 torque rises with voltage (one source: 1.2 → 1.6 kg·cm from 4.8 → 6 V; another lists 1.8 kg·cm at 4.8 V), but this board needs at least 6.1 V in: keep the pack above about 6.5 V
Capacitor C1	470–1000 µF, 10 V, low-ESR electrolytic	about 6.3 mm diameter × 11 mm	Absorbs the servo's current spikes so the regulator and the Mercury do not brown out
Arming	SPST switch ≥ 1 A, or a JST pull-pin / jumper	small slide switch	Keeps the servo dead while you handle it; reach it through the transition ring
Fuse (optional)	1.1 A resettable (polyfuse)	—	Limits a short; the LiPo can deliver far more than the wires should carry
Wire and plugs	24 AWG silicone for power, 28 AWG for the signal; JST-PH 2-pin on the battery (match the plug on the pack you buy)	—	Flexible, heat-tolerant, small

Estimated load: the servo idles near 10 mA and draws about 100–250 mA while moving; one deployment is under a second. A 300 mAh pack therefore lasts many hours of standby (estimate: the regulator's own draw I could not confirm).

2 Wiring schematic

V4 airbrake: servo power and signal (wiring schematic)



C1 = 470 - 1000 uF, 10 V, low-ESR, wired right at the servo pins. The regulator powers the servo; the Mercury only sends the position signal.

Ground is shared. Never join the servo red wire to a Mercury pin; confirm the Mercury servo-header pin order in its manual before wiring.

Servo power and signal. The Mercury has its own 50 mAh battery and does not power the servo.

How it is wired

- **Power path:** battery + → fuse → arming switch → regulator VIN; regulator VOUT (+5 V) → servo red wire and C1; regulator GND, C1 and the servo brown wire all return to the battery –.
- **Signal path:** one Mercury servo header (GP6 or GP7, set in the Mercury configuration as the air-brake servo channel) → servo orange wire. The Mercury GND goes to the same ground rail.
- **The one rule:** the servo's red wire takes power from the regulator only. If the Mercury servo header has a power pin, leave it unconnected (pull the red wire out of the Mercury-side connector shell, or use a servo lead with the red wire cut on that side). Two supplies tied together can damage the Mercury or the regulator.
- **Order of assembly:** bench-test with the servo and a 5 V supply first, then the regulator on the battery, with the switch off until the last step.

3 What I could not verify, and what it means for the CAD

The battery I recommend is bigger than the placeholder in the V4 bay. The sled's battery pocket is 40.8 × 18.8 × 10.9 mm; the Tattu is about 45–47 × 16–17 × 12–13 mm. It does not fit as modelled: the pocket would need to be about 48 × 18 × 13.6 mm, and the bay about 2.7 mm taller (housing about 97 mm instead of 94.3 mm). I have not changed the CAD yet. Tell me whether you want the bay enlarged for this pack, or a smaller pack (a 2S pack of 150–200 mAh would be thinner, but I could not verify any such pack's dimensions).

- **Mercury servo header:** the Altimeter Cloud flight page I found says the two onboard servo headers are GP6 and GP7, but I could not read the pin order or the signal voltage. Check its manual (the "Mercury V1 schematic" page). The SG90 usually accepts a 3.3 V signal when powered at 5 V, but test it on the bench before flying.
- **SG90 figures vary by source:** stall current 0.3 to 0.55 A, torque 1.2 to 1.8 kg·cm at 4.8 V depending on the version. The V3/V4 torque check used 0.176 N·m (1.8 kg·cm) at 5 V; measure your own servo.
- **Regulator input:** Pololu warns of voltage spikes above its 36 V maximum with long leads at high input voltages; with a 2S pack (8.7 V full) this is not a concern. The D24V10F5 needs at least 5.1 V in, so stop flying a 2S pack that has sagged near 6 V.
- **LiPo safety:** the HV pack charges to 4.35 V per cell and needs a LiHV-capable charger setting; do not discharge below about 3.3 V per cell; keep the pack away from the ejection-charge heat and from crushing; use a fire-safe bag for charging.
- **Altimeter pressure:** the Mercury reads static pressure, so the bay needs vent holes to the outside (not modelled in the CAD).

Sources

- Altimeter Cloud flight page (GP6/GP7 servo headers, air-brake channel) and the Mercury V1 schematic page (not readable from this session).
- Pololu D24V10F5 5 V, 1 A step-down regulator and D24V10F6 6 V version.
- Tattu 300 mAh 2S listings: GetFPV, Aloft Hobbies, Gens Ace; Turnigy nano-tech 2S 300 mAh: Botland.
- SG90 figures: SG90 datasheet copy and ProtoSupplies.